

A DEEP LEARNING FRAMEWORK FOR EARLY FOREST FIRE DETECTION USING COMPUTER VISION



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INTRODUCTION

Wildfire events have intensified globally, causing significant impacts to the environment and economy. In January 2025 alone, more than 12,000 structures were incinerated in Southern California. Despite ongoing advancements of technology, current detection methods remain limited.



One effective approach to solve this problem is by utilizing computer vision, an Artificial Intelligence subfield, which enables early and accurate fire detection.

METHODOLOGY

Data Collections and Preprocessing

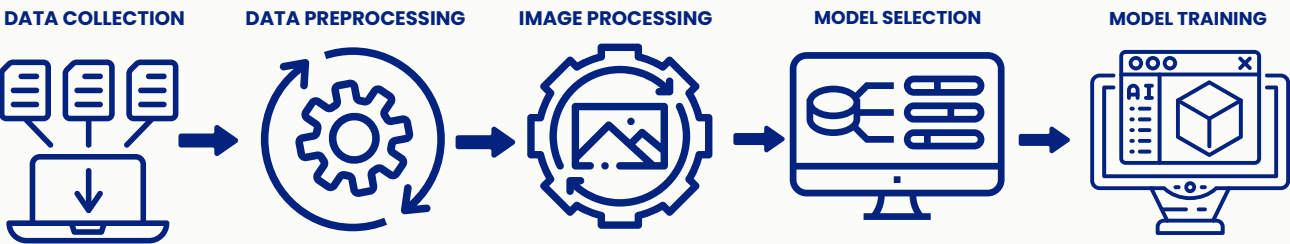
A dataset consisting of 2,509 fire and non-fire images was collected from Roboflow (2022). All images were preprocessed and resized to a resolution of 640 × 640 pixels to ensure consistency.

Image Processing

Image enhancement steps were applied to improve sharpness and reduce blur, to minimize the risk of incorrect detections or false positives.

Model Selection and Training

Model selection involved evaluating three different YOLOv8 variants which are, Nano (n), Small (s), and Medium (m). Each model was trained and tested on a system equipped with a dedicated GPU.



LITERATURE REVIEW

Based on previous work by Lei et al. (2024) proposed and demonstrated the practical advantages of employing a lightweight YOLOv8 model for forest fire detection. Their model achieved an accuracy of 92.3% for mAP@0.5 and real-time inference exceeding 30 FPS. However, with notable limitation of reduced effectiveness in detecting smaller scaled fires under poor lighting conditions.

References: L. Lei, R. Duan, F. Yang, and L. Xu, "Low complexity forest fire detection based on improved yolov8 network," Forests, vol. 15, no. 9, p. 1652, 2024.

OBJECTIVE

To implement and evaluate a deep learning-based system utilizing the YOLOv8 architecture for early forest fire detection, with an emphasis on:

- Develop a Cost-Effective YOLOv8 Architecture.
- Real-time detection capabilities (≥ 30 FPS).

CONCLUSION

The proposed YOLOv8-based framework demonstrates a highly effective balance of accuracy and computational efficiency, thus, making it cost effective or suitable for resource constrained environments. Each YOLOv8 model, n, m, and s has their own strength. The YOLOv8n model variant offers the best balance for real-time use, achieving an average of 58.4 FPS and high precision (0.961).

FUTURE WORKS

Expand the dataset to include a wider variety of fire and non-fire images under different environmental conditions, such as smoke, poor/low-light conditions, and others.

RESULTS AND ANALYSIS

This is the results from testing the models:

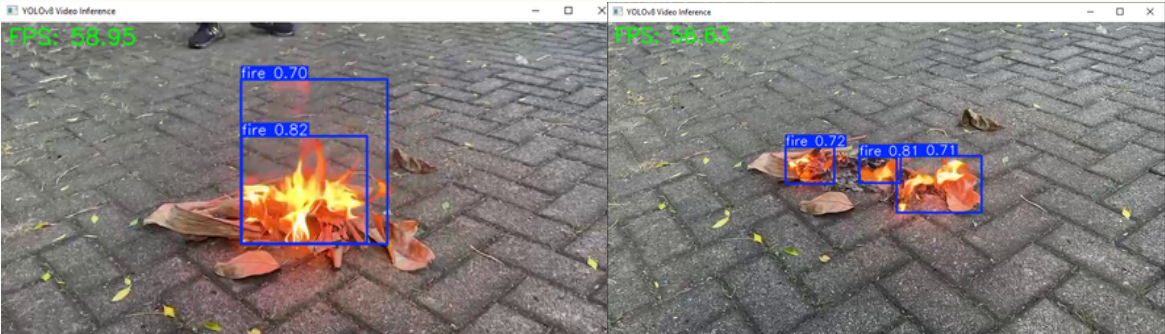
TABLE I
COMPARISON OF MODEL PERFORMANCE METRICS

Model	Precision	Recall	mAP@0.5	mAP@0.5-0.95	FPS
YOLOv8n	0.961	0.943	0.985	0.805	58.4
YOLOv8s	0.970	0.933	0.983	0.817	27.9
YOLOv8m	0.952	0.939	0.982	0.806	13.8

Table I compares the performance of the three YOLOv8 variants, which are nano (n), small (s), and medium (m) on the Roboflow Continuous Fire Detection dataset. Fire detection accuracy was evaluated using key metrics, including precision, recall, and mean average precision (mAP) at IoU thresholds 0.5 and 0.5-0.95.

The small model variant (YOLOv8s) achieved the highest mAP@0.5-0.95 with a score of 0.817, exhibiting superior ability to accurately detect and locating the fire. However, the larger YOLOv8m model, despite its increased parameter count, fails to surpass the smaller models in performance across all the key metrics due to overfitting or insufficient fine-tuning during training. Although the small model achieved the best mAP@0.5-0.95, the nano variant achieves best in recall, mAP@0.5 and FPS, which results in a better balance between efficiency and accuracy. Among the three models that were tested, the nano variant has a substantially lower parameter counts (3,151,904) than small (11,156,544) and medium variant (25,886,080).

This is the results from real-time prediction test (YOLOv8n Variant):



As demonstrated above, the YOLOv8 nano variant performs exceptionally well in real-time prediction, accurately detecting and locating fire instances with bounding boxes. Table I shows that the nano model achieves an average of 58.4 FPS (frames per second) during real-time inference, significantly outperforming the small and medium variants, which output lower FPS due to their higher parameter counts. This superior efficiency, and combined with its high precision and accuracy, makes the nano variant the preferred choice for practical use in resource constrained environments. Thus, the nano variant offers greater cost efficiency for smaller edge devices, like Raspberry Pi or Nvidia Jetson Nano.